IF3151 Advanced Data Structures and Algorithms
Mini Project
Topic: Support Ticketing System with Max Heap-based Prioritization

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Abstract

This abstract describes the implementation of a support ticketing system that internally uses a max heap to sort tickets created, based on the following attributes: Assigned Priority (P1/P2/P3) and the number of days till or past the assigned deadline.
It is intended to make it easier for support teams to decide which tickets to prioritize without having to guess which ones are more important to be addressed at a given moment.

Priority Score Formula

In order to sort the created tickets, their priority scores are computed using the following formula:
$Score = P + (10000 * e^{-D})$, where:

- P = Assigned priority for the ticket. From highest to lowest: P3 = 300, P2 = 200, P1 = 100.
- D = Number of days until the specified deadline. A negative value indicates that the ticket is overdue.

Application Functionality

- The application's Max Heap implementation is used to sort the tickets created in descending order of priority.
- SQLite is used for persistent storage, with changes to ticket entries written to it before recalculating priorities and recreating the priority list.
- The SQLite database will consist of a single table "Tickets", with the following columns:
  - uuid (String) – Unique Identifier
  - author (String) – Author of the ticket
  - description (String) – Actual text content of the ticket
  - creation_time (Integer) – UNIX timestamp denoting when the ticket was created.
  - deadline_time (Integer) – UNIX timestamp denoting when the ticket's resolution is due.
  - priority (Integer) – Static priority addend for computing the priority score, i.e. 300 for P3, 200 for P2, and 100 for P1.
- When computing the difference between deadline_time and the creation_time, the nearest day count is considered when computing the priority score. e.g. If the millisecond difference falls just short of one day, the number of days remaining is considered to be zero.